

Dongwook Kwon

Room 701, Saebit Hall, 20 Gwangun-ro, Nowon-gu, Seoul 01897, Republic of Korea

+82 10-6612-9246 | tomiyadragon@gmail.com | dongwooky.github.io | github/dongwooky | in/dongwooky

Experience

Suresoft Technologies

AI Engineer, Intelligent Agent Technology Team

Seongnam, Gyeonggi, Korea

Dec. 2025 – Present

- Developing an AI validation system utilizing AI agent-based architectures.
- Designing and implementing systems to verify and evaluate AI models through autonomous and collaborative AI agents.
- Building agentic AI systems to enable systematic, scalable, and reliable AI verification processes.

Bio-Computing and Machine Learning Laboratory, Kwangwoon University

Research Assistant

Seoul, Korea

Aug. 2021 – Dec. 2025

- Conducted research on deep learning algorithms, focusing on anomaly detection in time-series data, including Electrocardiogram (ECG) and Cyber-Physical Systems (CPS).
- Participated in the development of an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.

LG Electronics Canada

Research Project

Toronto, ON, Canada

Jan. 2025 – Jul. 2025

- Participating in the development of an assessment framework for agentic AI systems utilizing LLMs to evaluate performance.

Qualcomm Institute, University of California San Diego

AI Development Project, Summer Program

San Diego, CA, USA

Jul. 2022 – Aug. 2022

- Participated in the Personality-Based Drug Addiction Prediction System project.
- Attended seminars on deep learning and machine learning algorithms.

Education

Kwangwoon University

Master of Science in Computer Engineering

Seoul, Korea

Mar. 2024 – Feb. 2026

- GPA: 4.33/4.5 (Percentage: 98.1%)

University of Toronto

Graduate Research Program in Mechanical and Industrial Engineering

Toronto, ON, Canada

Jan. 2025 – Jul. 2025

- GPA: 3.68/4.0

Kwangwoon University

Bachelor of Science in Electronics and Communications Engineering

Seoul, Korea

Mar. 2019 – Feb. 2024

- GPA: 3.41/4.5 (Percentage: 88.1%)

Honors & Awards

Major Awards

- Aug. 2022* **Achievement Award**, Qualcomm Institute AI Development Project — UC San Diego *United States*
- Dec. 2021* **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project – Minister of Oceans and Fisheries *South Korea*

Academic Recognition

- Nov. 2025* **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers *South Korea*
- Nov. 2024* **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024) *South Korea*
- Nov. 2023* **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering *South Korea*

Competitions

- Dec. 2025* **Silver Award**, 23rd Embedded Software Competition (KIISE) *South Korea*
- Dec. 2022* **Bronze Award**, Hanium ICT Mentoring Competition – FKII *South Korea*
- Sep. 2022* **Excellence Award**, 18th Kwangwoon ICT Exhibition (KWIX) — Kwangwoon University *South Korea*
- Oct. 2021* **Grand Prize**, MY (Multi-Y) Capstone Design Competition — Kwangwoon University *South Korea*

Projects

Agentic AI Evaluation Framework

LG Electronics Canada

Toronto, ON, Canada
Jan. 2025 – Present

- Developed an evaluation framework for agentic AI systems using LLMs to assess performance.

AI-based Intrusion Detection and Attack Packet Crafting Technologies for APR1400

Sejong University

Seoul, Korea
Jul. 2022 – Present

- Developed an AI defense system to protect nuclear power plants from cyber attacks.

Quadruped Robot Using Vision SLAM with a Depth Camera

Ministry of Science and ICT

Seoul, Korea
Apr. 2022 – Nov. 2022

- Developed an autonomous quadruped robot using a navigation algorithm with SLAM and a depth camera.

Active Kill-Switch and Biomarker-Based Defense for Life-Threatening IoT Medical Devices

Kookmin University

Seoul, Korea
Sep. 2021 – Dec. 2022

- Developed a security system for heart implant devices using an anomaly detection algorithm.

Development of a Smart Logistics System Using Computer Vision

Ministry of Oceans and Fisheries

Seoul, Korea
Apr. 2021 – Nov. 2021

- Developed a logistics detection system incorporating location tracking and stack layer identification for containers.

Publications

Journal Articles

- Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, "Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise," *Electronics*, vol. 14, no. 5, p. 890, 2025.

- **D. Kwon**, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, "Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments," under review at *Data Mining and Knowledge Discovery* (Springer Nature), 2025.

Conference Papers

- R. Chang[†], **D. Kwon**[†], J. Lee[†], and N. Verma, "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) – Industry Track (Poster)*, 2026. [†]Equal contribution
- J. Lee[†], R. Chang[†], **D. Kwon**[†], H. Singh, and N. Verma, "GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems," in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, 2025. [†]Equal contribution
- **D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park, "Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers," in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. — **Best Poster Award – Silver Winner**
- **D. Kwon**, Y. Kang, and C. Park, "Enhancing medical device security with GNN-GRU anomaly detection model," in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. — **Outstanding Paper Award**

Patents

- **Issued Patent**, "Method for Implantable Medical Device to Detect Anomaly in Real Time," Patent No. KR 10-2961984, Issued on April 30, 2026.
- **Issued Patent**, "GPT API-Based Network Anomaly Detection," Patent No. KR 10-2848814, Issued on August 18, 2025.

Leadership Experience

- **President**, Kwangwoon International Student Association — Lead intercultural programs and student engagement initiatives.
- **Director of Filming and Editing**, Kwangwoon Broadcasting Center (KWBC) — Oversaw media production and content delivery.
- **Class Representative**, Dept. of Electronics and Communications Engineering — Represented student interests in academic affairs.
- **Executive Member**, Amateur Radio Club — Organized radio workshops and technical exhibitions.